

Name: Manahil Habib

Section: 5-B

Reg no: 2023-BSE-042

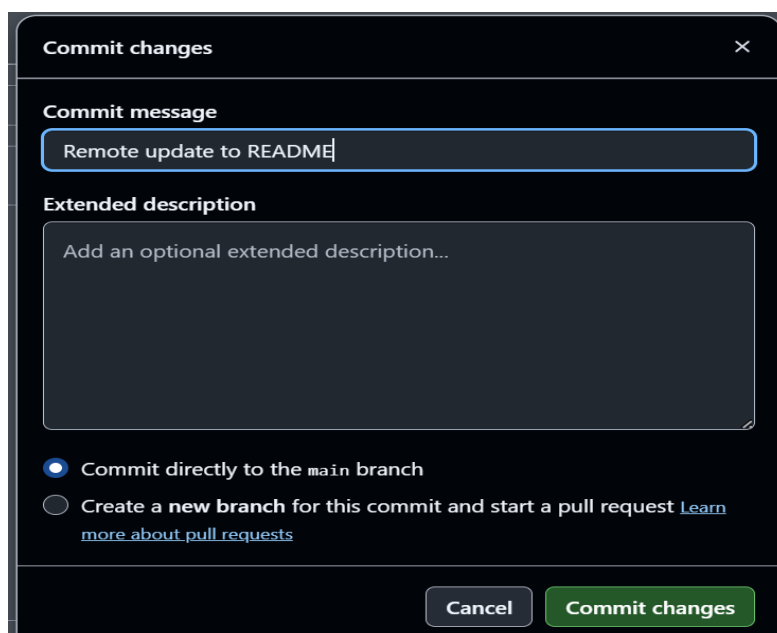
Lab 3

Task1

1: Open your repository on **GitHub** and edit the README.md file directly in the browser.



```
1 # cc-manahilhabib-042
2 # assignment1
3
4 **Student Name:** Manahil Habib
5 **Roll Number:** 042
6
7 My first Gitea repo running inside Codespaces
8
9 This line was added remotely from GitHub.
10 |
11
```



Commit changes [X]

Commit message

Remote update to README

Extended description

Add an optional extended description...

☒ Commit directly to the main branch

☐ Create a new branch for this commit and start a pull request [Learn more about pull requests](#)

Cancel Commit changes

2: On your **local machine**, open the same repository folder.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git status
git branch
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
* main

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ notepad README.md

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add README.md
git commit -m "Local update to README"
[main dea5887] Local update to README
1 file changed, 7 insertions(+)

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ |
```

3: Try to push:

```
! [rejected] main -> main (fetch first)
error: failed to push some refs to 'github.com:yourusername/cc-manahilhabib-042.git'
hint: Updates were rejected because the remote contains work that you do not have
```

4: To fix this, pull the latest changes from remote:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git pull --no-rebase origin main
From https://github.com/manahilhabib031/cc-manahilhabib-042
 * branch          main          -> FETCH_HEAD
Already up to date.
```

5: Push again:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 330 bytes | 330.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
 c229751..dea5887  main -> main
```

6: Now repeat the same process, but this time use **rebase instead of merge**:

- Make another remote edit on GitHub in README.md.
- Then make another local edit and commit it.

- Instead of using git pull, run

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ notepad README.md

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add README.md
git commit -m "Local update for rebase test"
[main ac786fc] Local update for rebase test
1 file changed, 2 insertions(+)

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 953 bytes | 56.00 KiB/s, done.
From https://github.com/manahilhabib031/cc-manahilhabib-042
* branch      main      -> FETCH_HEAD
   dea5887..1ef8700 main  -> origin/main
Successfully rebased and updated refs/heads/main.
```

Push again:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 302 bytes | 151.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
   1ef8700..59c865e main -> main
branch 'main' set up to track 'origin/main'.
```

Task 2

- 1: On **GitHub (remote)**, open your README.md file and change an existing line.

cc-manahilhabib-042

assignment1

Student Name: Manahil Habib

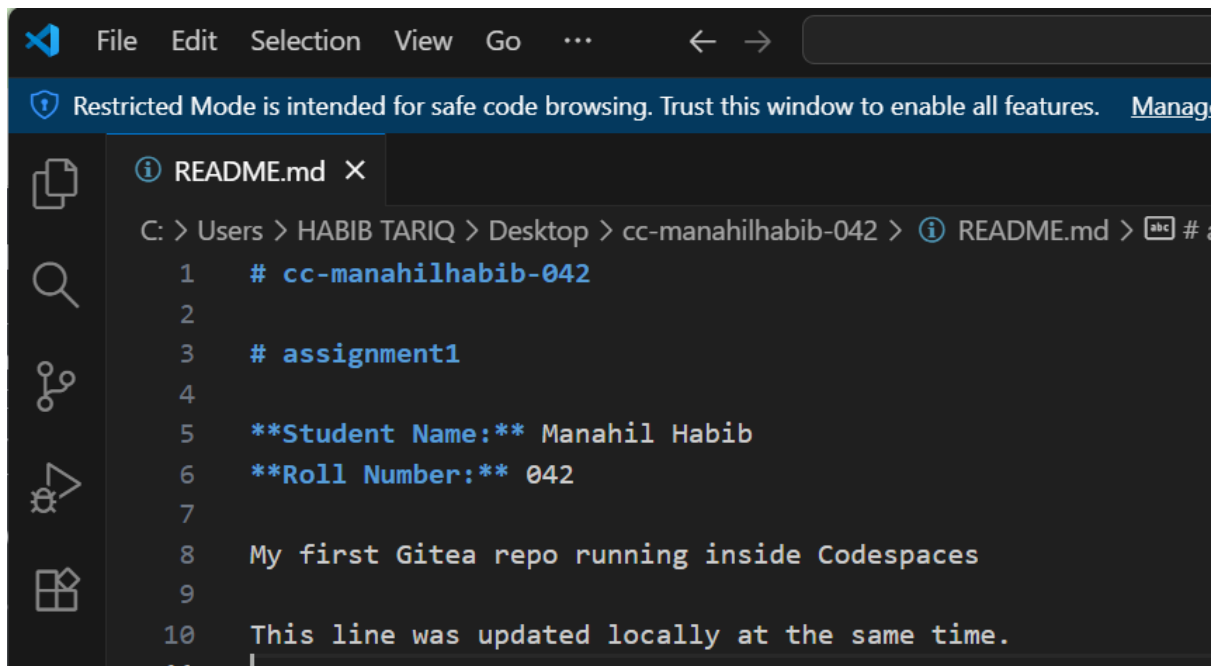
Roll Number: 042

My first Gitea repo running inside Codespaces

This line was updated remotely again.

2: On your **local machine**, edit the **same line** in the same file but make a **different change**:

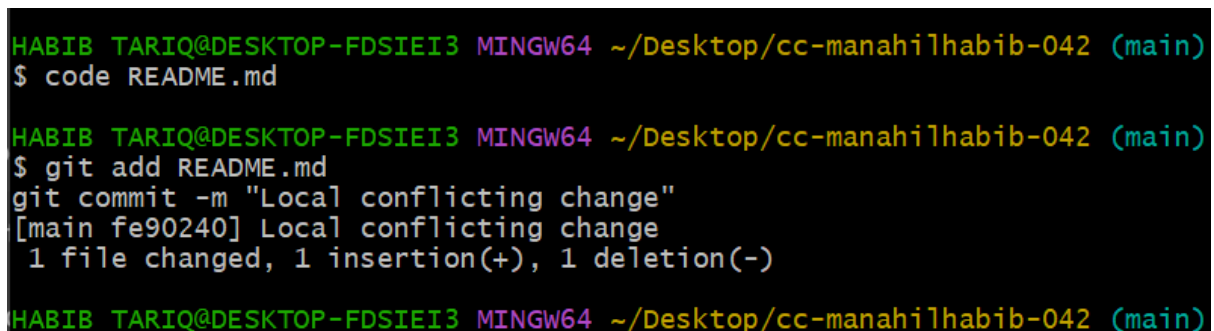
g



The screenshot shows a code editor window with a dark theme. The title bar indicates 'Restricted Mode is intended for safe code browsing. Trust this window to enable all features. Manage'. The file explorer on the left shows 'README.md' selected. The main editor area displays the following content:

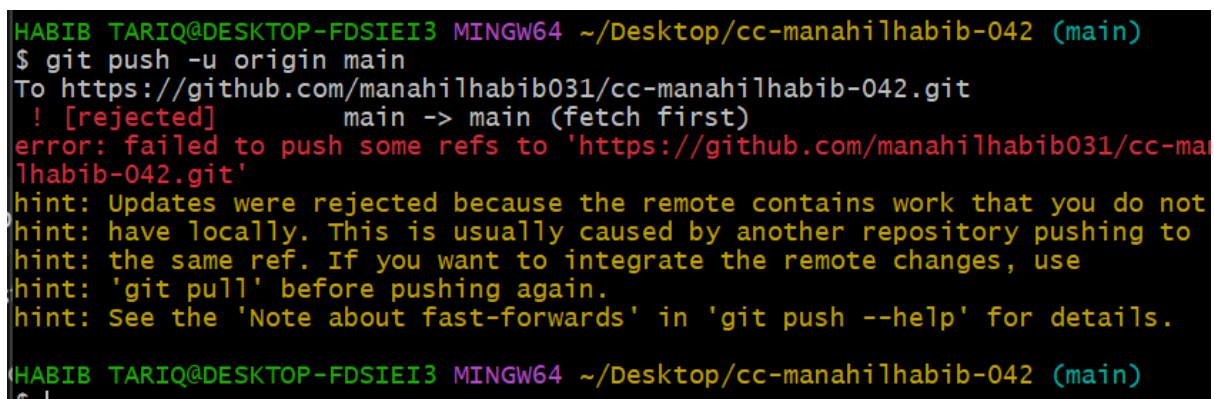
```
C: > Users > HABIB TARIQ > Desktop > cc-manahilhabib-042 > README.md > #  
1  # cc-manahilhabib-042  
2  
3  # assignment1  
4  
5  **Student Name:** Manahil Habib  
6  **Roll Number:** 042  
7  
8  My first Gitea repo running inside Codespaces  
9  
10 This line was updated locally at the same time.  
11
```

3: Stage and commit your local change:



```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)  
$ code README.md  
  
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)  
$ git add README.md  
git commit -m "Local conflicting change"  
[main fe90240] Local conflicting change  
1 file changed, 1 insertion(+), 1 deletion(-)  
  
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
```

4: Try to push:



```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)  
$ git push -u origin main  
To https://github.com/manahilhabib031/cc-manahilhabib-042.git  
! [rejected]        main -> main (fetch first)  
error: failed to push some refs to 'https://github.com/manahilhabib031/cc-manahilhabib-042.git'  
hint: Updates were rejected because the remote contains work that you do not  
hint: have locally. This is usually caused by another repository pushing to  
hint: the same ref. If you want to integrate the remote changes, use  
hint: 'git pull' before pushing again.  
hint: See the 'Note about fast-forwards' in 'git push --help' for details.  
  
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)  
$
```

5: Pull with rebase to bring in remote changes:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 956 bytes | 47.00 KiB/s, done.
From https://github.com/manahilhabib031/cc-manahilhabib-042
* branch          main          -> FETCH_HEAD
   59c865e..b0e68ab main        -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply fe90240... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase -
abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply fe90240... # Local conflicting change
```

6: Open the README.md file in your editor — you'll see conflict markers like this:

```
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<< HEAD (Current Change)
This line was updated both locally and remotely (merged version).
This line was updated remotely again.

=====
This line was updated locally at the same time.
>>>>>> fe90240 (Local conflicting change) (Incoming Change)
```

7: After fixing the file, mark the conflict as resolved and continue the rebase:

```
[detached HEAD 9643f1d] Local conflicting change
1 file changed, 4 insertions(+)
Successfully rebased and updated refs/heads/main.

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
```

8: Finally, push your changes:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 403 bytes | 201.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
   b0e68ab..9643f1d main -> main
branch 'main' set up to track 'origin/main'.
```

Task 3

1: Create a new folder named textfiles inside your repository:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ cd "/c/Users/HABIB TARIQ/Desktop/cc-manahilhabib-042"
mkdir textfiles
echo "File A content" > textfiles/a.txt
echo "File B content" > textfiles/b.txt
echo "File C content" > textfiles/c.txt
```

2: Inside the textfiles folder, create three text files:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ cd "/c/Users/HABIB TARIQ/Desktop/cc-manahilhabib-042"
mkdir textfiles
echo "File A content" > textfiles/a.txt
echo "File B content" > textfiles/b.txt
echo "File C content" > textfiles/c.txt

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add .
git commit -m "Added textfiles directory with three files"
git push origin main
warning: in the working copy of 'textfiles/a.txt', LF will be replaced by CRLF t
he next time Git touches it
warning: in the working copy of 'textfiles/b.txt', LF will be replaced by CRLF t
he next time Git touches it
warning: in the working copy of 'textfiles/c.txt', LF will be replaced by CRLF t
he next time Git touches it
[main ca3fe8f] Added textfiles directory with three files
3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 471 bytes | 117.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
9643f1d..ca3fe8f main -> main
```

3: Add and commit the new directory:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ echo "textfiles/*" > .gitignore

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add .gitignore
git commit -m "Added .gitignore to ignore textfiles directory"
git push origin main
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main 03b098a] Added .gitignore to ignore textfiles directory
1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 358 bytes | 179.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
ca3fe8f..03b098a main -> main
```

4: Now, create a .gitignore file in the root of your repository:

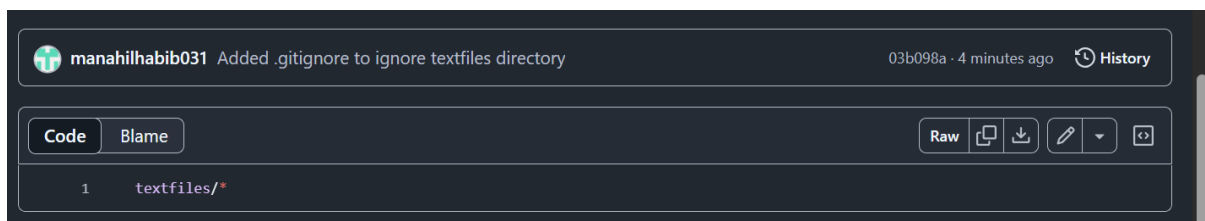
```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ echo "textfiles/*" > .gitignore
```

5: Add and commit the .gitignore file:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add .gitignore
git commit -m "Added .gitignore to ignore textfiles directory"
git push origin main
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
on branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
Everything up-to-date
```

6: Go to **GitHub** and check your repository — notice that the textfiles directory is **still visible** on the remote.



7: To remove already tracked files from Git (but not from your local system), run:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
```

8: Commit and push again:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add .
git commit -m "Removed tracked textfiles directory"
git push origin main
[main 50c8e2b] Removed tracked textfiles directory
3 files changed, 3 deletions(-)
delete mode 100644 textfiles/a.txt
delete mode 100644 textfiles/b.txt
delete mode 100644 textfiles/c.txt
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 246 bytes | 123.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
03b098a..50c8e2b main -> main
```

9: Check your GitHub repository again — the textfiles folder should now be **deleted remotely**.



Task 4

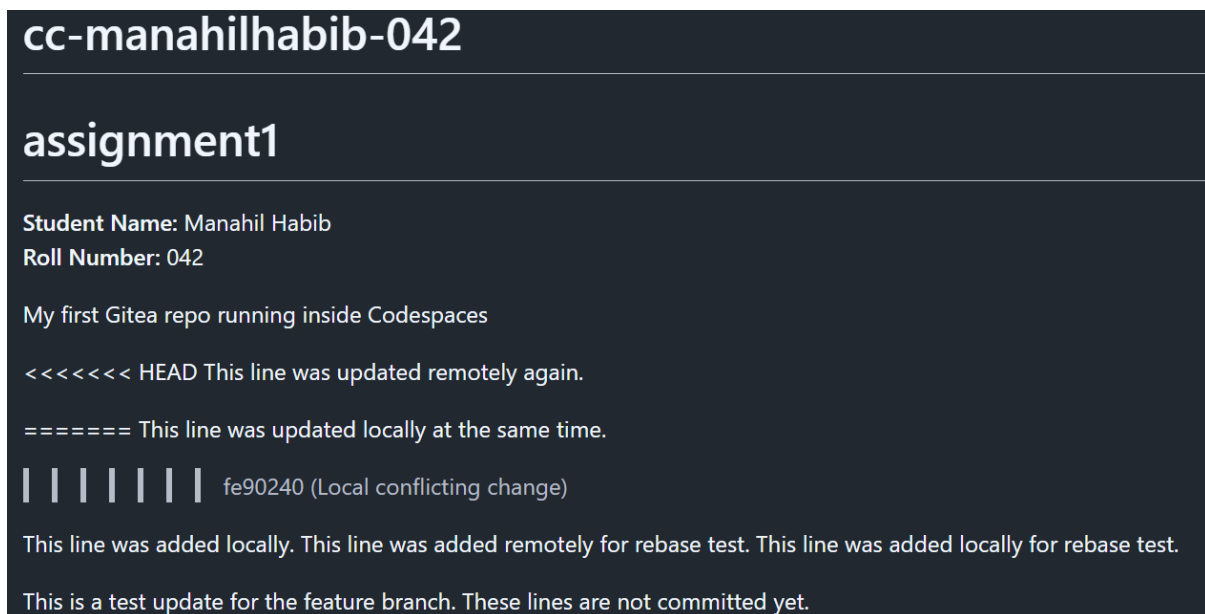
1: Create a feature-branch.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
```

2: Modify any file (for example, README.md) by adding a few test lines.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (feature-branch)
$ git add README.md
git commit -m "Changes the README.md file"
On branch feature-branch
nothing to commit, working tree clean
```

3: Modify any file (for example, README.md) by adding a few new lines.



4: Without committing or stashing the changes, try to switch to another branch:

```
error: Your local changes to the following files would be overwritten by checkout
README.md
Please commit your changes or stash them before you switch branches.
```


5: To fix this, **stash your changes**:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~  
$ git stash  
fatal: not a git repository (or any of the parent directories): .git  
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~
```

6: Try switching branches again — it should now work:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)  
$ git checkout main  
Already on 'main'  
Your branch is up to date with 'origin/main'.
```

7: Return to your previous branch (for example, feature-branch):

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)  
$ git checkout feature-branch  
Switched to branch 'feature-branch'
```

8: Check your working directory status:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (feature-branch)  
$ git status  
On branch feature-branch  
nothing to commit, working tree clean  
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (feature-branch)  
$ nothing to commit, working tree clean  
bash: nothing: command not found
```

9: Now restore your stashed changes:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (feature-branch)  
$ git stash pop  
No stash entries found.
```

Task 5

1: View commit history:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git log --oneline
50c8e2b (HEAD -> main, origin/main, origin/HEAD, feature-branch) Removed tracked
  textfiles directory
03b098a Added .gitignore to ignore textfiles directory
ca3fe8f Added textfiles directory with three files
9643f1d Local conflicting change
b0e68ab Remote conflicting change
59c865e Local update for rebase test
1ef8700 Remote update for rebase test
dea5887 Local update to README
c229751 Remote update to README
0a23f26 Added name and roll number
fd5bd26 Create README.md
```

2: Copy any previous commit hash (e.g., a1b2c3d).

3: Checkout that commit:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git checkout 50c8e2b
Note: switching to '50c8e2b'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

    git switch -c <new-branch-name>

Or undo this operation with:

    git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 50c8e2b Removed tracked textfiles directory
```

4: To return to your main branch:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 ((50c8e2b...))
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

Task 6

1: Add a new line in any file and commit it:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ nano README.md

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add .
git commit -m "Added test line"
[main 2fce9b8] Added test line
1 file changed, 1 insertion(+), 1 deletion(-)
```

2: Add another change and commit again:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ nano README.md

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add .
git commit -m "Second test commit"
[main b101262] Second test commit
1 file changed, 2 insertions(+)
```

3: View the history before reset:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git log --oneline
b101262 (HEAD -> main) Second test commit
2fce9b8 Added test line
c61f30c (origin/main, origin/HEAD) Update README.md
6067b56 Update branch
50c8e2b (feature-branch) Removed tracked textfiles directory
03b098a Added .gitignore to ignore textfiles directory
ca3fe8f Added textfiles directory with three files
9643f1d Local conflicting change
b0e68ab Remote conflicting change
59c865e Local update for rebase test
1ef8700 Remote update for rebase test
dea5887 Local update to README
c229751 Remote update to README
0a23f26 Added name and roll number
fd5bd26 Create README.md
```

4: Check file contents for both changes:

This is the first test line. This is the second test line.

5: Perform a soft reset (keeps changes in working directory):

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git reset --soft HEAD~1
```

6: Check commit history after soft reset:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git log --oneline
2fce9b8 (HEAD -> main) Added test line
c61f30c Update README.md
6067b56 Update branch
50c8e2b (feature-branch) Removed tracked textfiles directory
03b098a Added .gitignore to ignore textfiles directory
ca3fe8f Added textfiles directory with three files
9643f1d Local conflicting change
b0e68ab Remote conflicting change
59c865e Local update for rebase test
1ef8700 Remote update for rebase test
dea5887 Local update to README
c229751 Remote update to README
0a23f26 Added name and roll number
fd5bd26 Create README.md
```

7: Verify changes in the file:

assignment1

Student Name: Manahil Habib

Roll Number: 042

My first Gitea repo running inside Codespaces

<<<<<< HEAD This line was updated remotely again.

==== This line was updated locally at the same time.

| | | | | | fe90240 (Local conflicting change)

This line was added locally. This line was added remotely for rebase test. This line was added remotely for rebase test.

This is a test update for the feature branch. These lines are not committed yet.

This is the first test line. This is the second test line.

8: Check git status:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

9: Perform commit

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git commit -m "Second Test commit"
[main e936061] Second Test commit
 1 file changed, 2 insertions(+)
```

10: Perform a hard reset (discards changes completely):

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git reset --hard HEAD~1
HEAD is now at 2fce9b8 Added test line
```

11: Check commit history after hard reset:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git log --oneline
2fce9b8 (HEAD -> main) Added test line
c61f30c Update README.md
6067b56 Update branch
50c8e2b (feature-branch) Removed tracked textfiles directory
03b098a Added .gitignore to ignore textfiles directory
ca3fe8f Added textfiles directory with three files
9643f1d Local conflicting change
b0e68ab Remote conflicting change
59c865e Local update for rebase test
1ef8700 Remote update for rebase test
dea5887 Local update to README
c229751 Remote update to README
0a23f26 Added name and roll number
fd5bd26 Create README.md
```

12: Verify changes in the file:

This is the first test line.

13: Check git status:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean
```

Task 7

1: Make a small change in any file.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ nano README.md
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
```

2: Stage it and commit:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git add .
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git commit -m "Fix log message"
[main d403a27] Fix log message
1 file changed, 2 insertions(+)
```

3: Realize you forgot to change another file. Update it now.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ nano README.md
```

4: Add it and amend the last commit:

```
[main 53ef3f8] Fix log message
Date: Wed Oct 15 11:35:43 2025 +0500
1 file changed, 3 insertions(+)
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$
```

Task 8

1: Make a change and commit it:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ echo "temporary text" >> temp.txt
git add .
git commit -m "Added temporary text file"
warning: in the working copy of 'temp.txt', LF will be replaced by CRLF the next time Git touches it
[main 3a96dde] Added temporary text file
1 file changed, 1 insertion(+)
create mode 100644 temp.txt
```

2: Now revert that commit safely (do not delete it):

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git log --oneline
git revert <commit-hash>
4c65c11 (HEAD -> main, origin/main, origin/HEAD) Added temporary text file
066d451 Fix log message
3f19586 Update README.md
b101262 Second test commit
2fce9b8 Added test line
c61f30c Update README.md
6067b56 Update branch
50c8e2b (feature-branch) Removed tracked textfiles directory
03b098a Added .gitignore to ignore textfiles directory
ca3fe8f Added textfiles directory with three files
9643f1d Local conflicting change
b0e68ab Remote conflicting change
59c865e Local update for rebase test
1ef8700 Remote update for rebase test
dea5887 Local update to README
c229751 Remote update to README
0a23f26 Added name and roll number
fd5bd26 Create README.md
bash: syntax error near unexpected token `newline'
```

3: Push the revert commit:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git push origin main
Everything up-to-date
```

Task 9

1: Create a new branch:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (main)
$ git checkout -b test-force
Switched to a new branch 'test-force'
```

2: Make and commit a small change.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (test-force)
$ git add .
git commit -m "Force push test commit"
[test-force 789111b] Force push test commit
1 file changed, 1 insertion(+)
```

3: Push it:


```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (test-force)
$ git push origin test-force
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 383 bytes | 191.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:   https://github.com/manahilhabib031/cc-manahilhabib-042/pull/new/test-force
remote:
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
 * [new branch]      test-force -> test-force
```

4: Perform a hard reset:

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (test-force)
$ git reset --hard HEAD~1
HEAD is now at 4c65c11 Added temporary text file
```

5: Push again

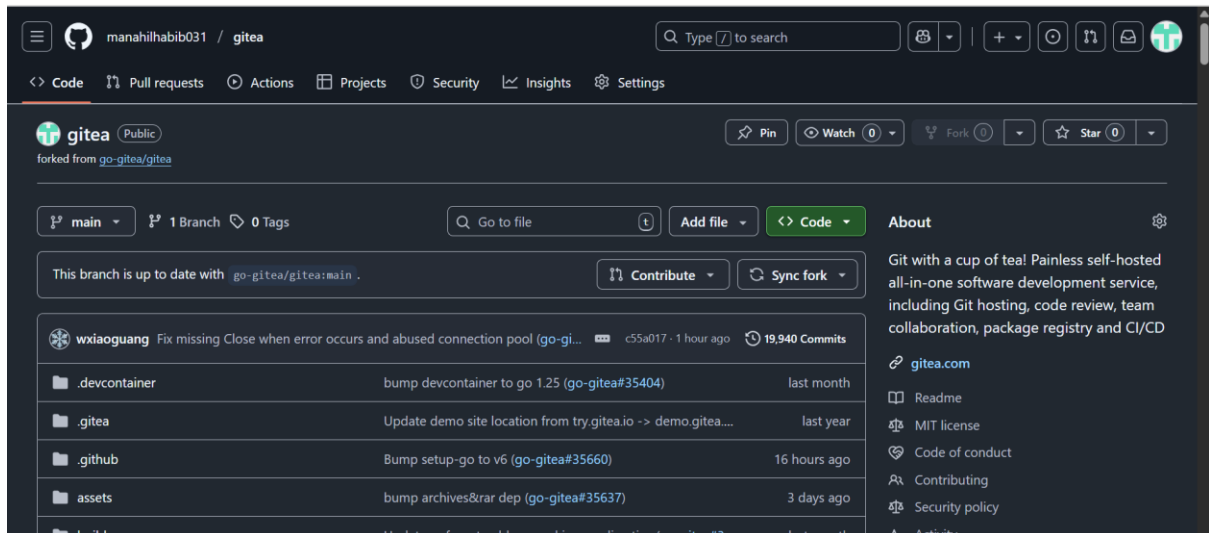
```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (test-force)
$ git push origin test-force
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
 ! [rejected]          test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'https://github.com/manahilhabib031/cc-manahilhabib-042.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

6: Now force-push to remote:

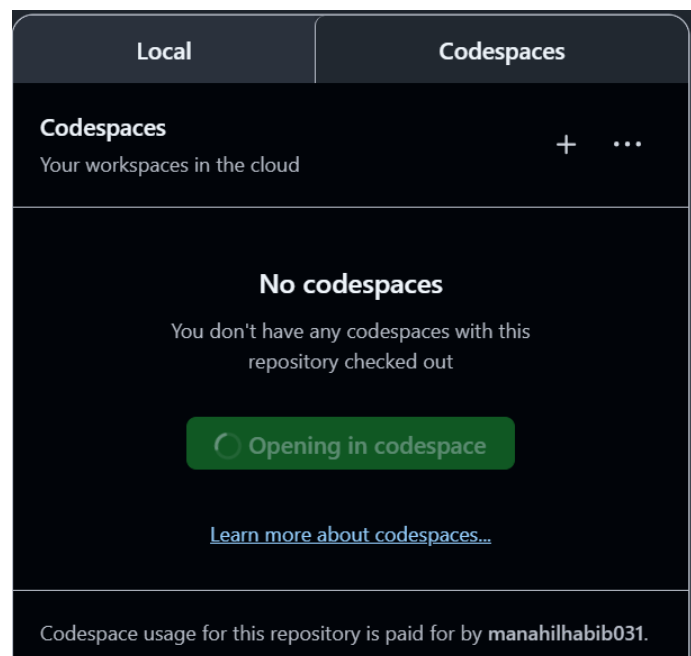
```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/cc-manahilhabib-042 (test-force)
$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/cc-manahilhabib-042.git
 + 789111b...4c65c11 test-force -> test-force (forced update)
```

Task 10

1: Fork the Gitea Repository



2: Open the Forked Repo in GitHub Codespaces



3: Start Gitea with Docker Compose

```
@manahilhabib031 →/workspaces/cc-manahilhabib-042 (main) $ docker --version
Docker version 28.3.1-1, build 38b7060a218775811da953650d8df7d492653f8f
@manahilhabib031 →/workspaces/cc-manahilhabib-042 (main) $ docker run -d --name=gitea \
  -p 3000:3000 \
  gitea/gitea:latest
Unable to find image 'gitea/gitea:latest' locally
latest: Pulling from gitea/gitea
9824c27679d3: Pull complete
307c463a9ebe: Pull complete
89986d54bad9: Pull complete
```

4: Access Gitea Web Interface

legendary-sniffle-pj5r56rjq4wf6pr6-3000.app.github.dev

Initial Configuration

If you run Gitea inside Docker, please read the [documentation](#) before changing any settings.

Database Settings

Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).

Database Type *

Path *

File path for the SQLite3 database.
Enter an absolute path if you run Gitea as a service.

General Settings

Site Title *

You can enter your company name here.

Repository Root Path *

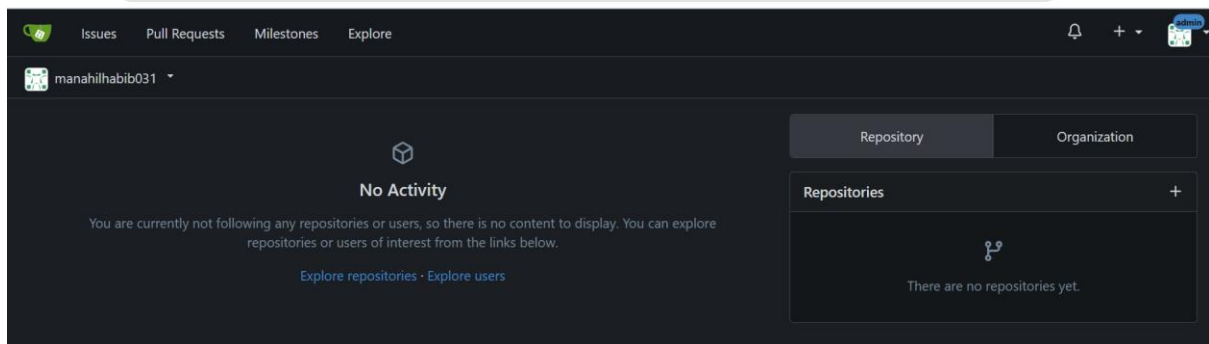
Remote Git repositories will be saved to this directory.

Git LFS Root Path

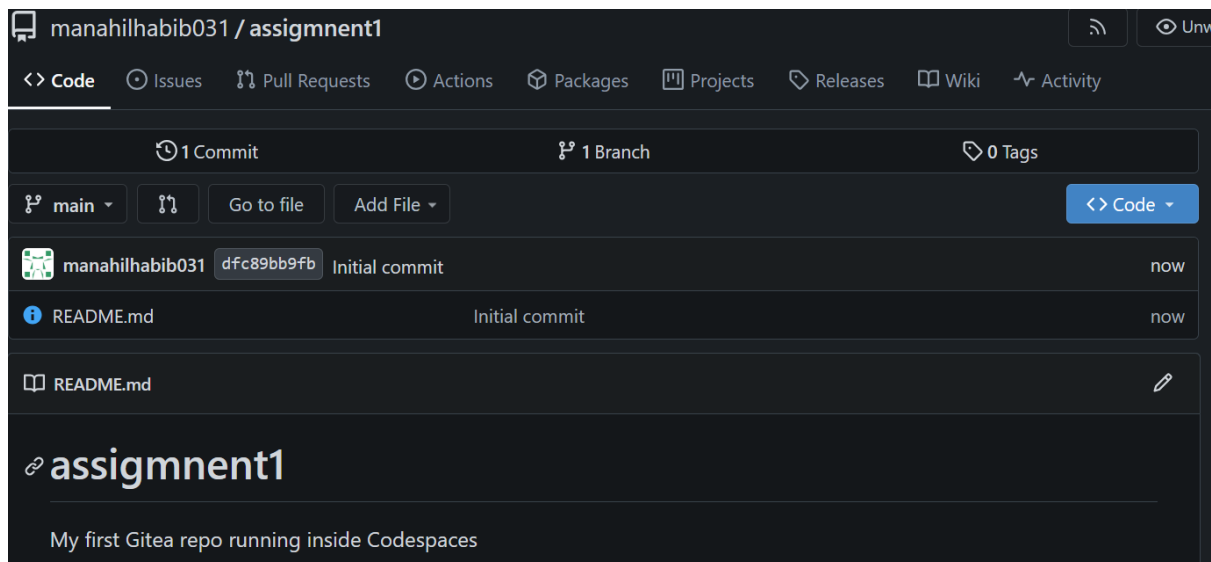
Files tracked by Git LFS will be stored in this directory. Leave empty to disable.

5: Install Gitea

6: Log In to Gitea

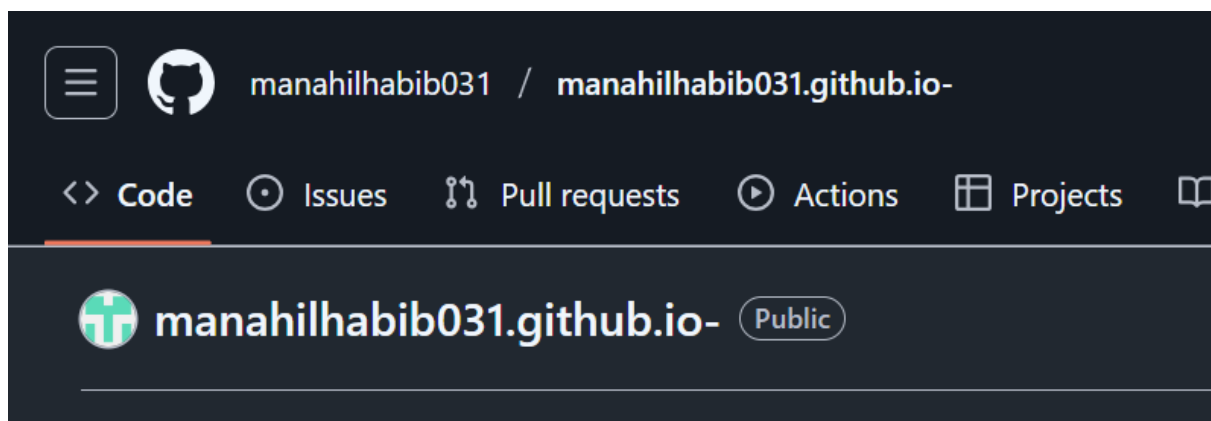


7: Create a New Repository in Gitea

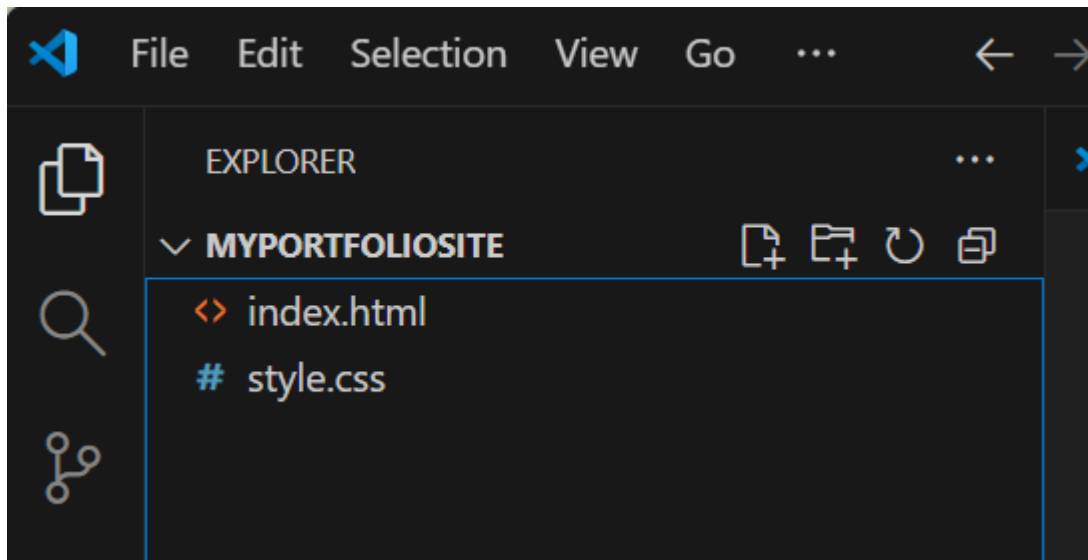


Task 11

1: Create a GitHub Pages Repository



2: Add Static Website Code

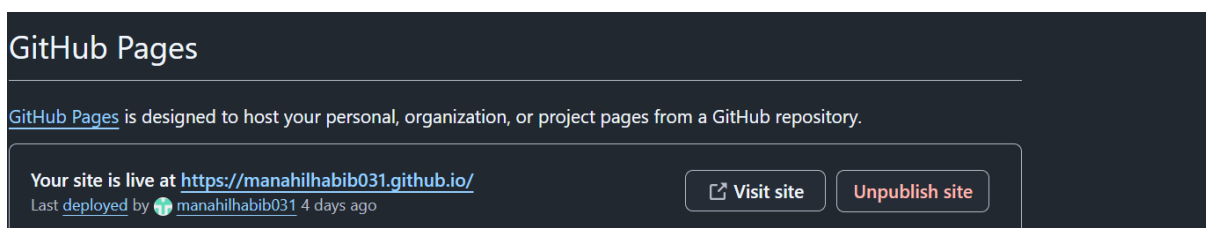


3: Push the Files to GitHub

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git remote add origin https://github.com/manahilhabib031/manahilhabib031.github.io-.git

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 2.65 KiB | 339.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/manahilhabib031.github.io-.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

4: Check GitHub Pages Settings



5: Visit Your Live Site

Manahil Habib

Software engineering Student | Developer | Tech Enthusiast

About Me

Hello! I'm **Manahil Habib**, a passionate Software engineering student who loves exploring coding, AI, and web development. I enjoy creating smart, clean, and efficient solutions through programming and design.

Education

BS Software engineering

Fatima Jinnah Women University – 2023-2027

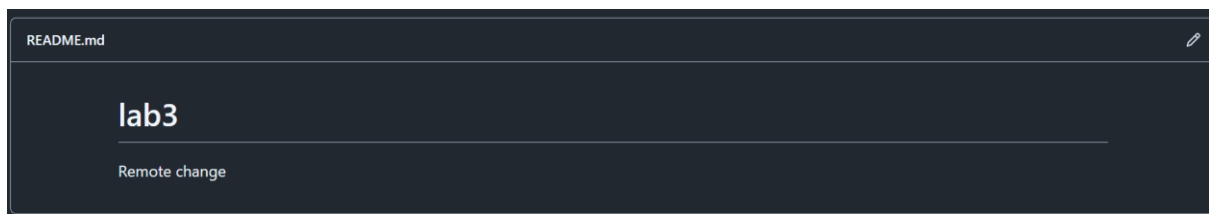
<https://manahilhabib031.github.io/>

Exam Evaluation Questions

1. Local vs Remote Conflict Resolution

Steps:

1. On GitHub, edit a file (e.g., README.md) and commit the change.



2. On your local machine, edit the same file differently (Avoid Conflict) and commit.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git add .
git commit -m "Local edit"
warning: in the working copy of 'README.md', LF will be replaced by CRLF the
t time Git touches it
[main 3014a5d] Local edit
1 file changed, 1 insertion(+)
create mode 100644 README.md
```

3. Try to push your local commit and observe the error.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git push origin main
fatal: unable to access 'https://github.com/manahilhabib031/manahilhabib031.github.io-.git/': Could not resolve host: github.com
```

4. Resolve the conflict by running git pull (merge), then push.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git pull origin main
From https://github.com/manahilhabib031/manahilhabib031.github.io-
 * branch                main                -> FETCH_HEAD
Already up to date.
```

5. Repeat with another remote/local change, but resolve using git pull --rebase, then push.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 335 bytes | 83.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/manahilhabib031.github.io-.git
 7e9ae9a..3014a5d main -> main
```

2. Manual Merge Conflict Handling

Steps:

1. On GitHub, change a specific line in a file and commit.

lab3

Remote change Welcome to my portfolio.

2. Locally, change the same line differently and commit.


```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ nano README.md

HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git add .
git commit -m "Local conflicting edit"
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
[main b32ae05] Local conflicting edit
1 file changed, 1 insertion(+)
```

3. Try to push your local change and observe the conflict error.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 380 bytes | 190.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/manahilhabib031.github.io-.git
3014a5d..b32ae05  main -> main
```

4. Use git pull --rebase to fetch changes and trigger the conflict.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git pull --rebase origin main
From https://github.com/manahilhabib031/manahilhabib031.github.io-
* branch          main          -> FETCH_HEAD
Already up to date.
```

5. Edit the conflicted file to resolve the conflict manually.
6. Mark the conflict resolved (git add <file>, git rebase --continue) and push.

3. Managing Ignored and Tracked Files

Steps:

1. Create a new folder (e.g., DocFiles) and add several files inside.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ mkdir DocFiles
cd DocFiles
echo "This is file 1" > file1.txt
echo "This is file 2" > file2.txt
echo "This is file 3" > file3.txt
cd
```

2. Commit and push the folder/files to GitHub.

```
HABIB TARIQ@DESKTOP-FDSIEI3 MINGW64 ~/Desktop/MyPortfolioSite (main)
$ git add .
git commit -m "Add DocFiles folder"
git push origin main
warning: in the working copy of 'DocFiles/file1.txt', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'DocFiles/file2.txt', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'DocFiles/file3.txt', LF will be replaced by CRLF
the next time Git touches it
[main e7144f5] Add DocFiles folder
3 files changed, 3 insertions(+)
create mode 100644 DocFiles/file1.txt
create mode 100644 DocFiles/file2.txt
create mode 100644 DocFiles/file3.txt
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 519 bytes | 173.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/manahilhabib031/manahilhabib031.github.io-.git
b32ae05..e7144f5 main -> main
```

3. Add the folder to your .gitignore file.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q3)
$ echo "DocFiles/" >> .gitignore
```

4. Commit and push the .gitignore update.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q3)
$ git add .gitignore
git commit -m "Added DocFiles to .gitignore"
git push origin exam-q3
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time
it touches it
[exam-q3 de548e1] Added DocFiles to .gitignore
1 file changed, 1 insertion(+)
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 357 bytes | 178.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:toobashafique065/lab3.git
1a67cfb..de548e1 exam-q3 -> exam-q3
```

5. Remove the folder from tracking using git rm -r --cached <folder>.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q3)
$ git rm -r --cached DocFiles
rm 'DocFiles/file1.txt'
rm 'DocFiles/file2.txt'
rm 'DocFiles/file3.txt'
```

6. Commit and push the change, then verify the folder is no longer tracked on GitHub.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q3)
$ git commit -m "Untracked DocFiles folder"
git push origin exam-q3
[exam-q3 36bf6d3] Untracked DocFiles folder
3 files changed, 3 deletions(-)
delete mode 100644 DocFiles/file1.txt
delete mode 100644 DocFiles/file2.txt
delete mode 100644 DocFiles/file3.txt
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 292 bytes | 292.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:toobashafique065/lab3.git
de548e1..36bf6d3 exam-q3 -> exam-q3
```

4. Commit History Manipulation and Recovery

Steps:

1. Make a change and commit it.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q4)
$ git add .
git commit -m "First change for Q4"
[exam-q4 c9657f3] First change for Q4
1 file changed, 1 insertion(+), 5 deletions(-)
```

2. Make another change and commit again.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q4)
$ git add .
git commit -m "Second change for Q4"
[exam-q4 1136fe9] Second change for Q4
1 file changed, 3 insertions(+), 1 deletion(-)
```

3. View your commit history.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q4)
$ git log --oneline
1136fe9 (HEAD -> exam-q4) Second change for Q4
c9657f3 First change for Q4
36bf6d3 (origin/exam-q4, origin/exam-q3, exam-q3) Untracked DocFiles folder
de548e1 Added DocFiles to .gitignore
1a67cfb Added DocFiles folder with files
1cb9f9b (origin/exam-q2, exam-q2) Q2 local conflict edit
42faa2d Q2 remote conflict edit
eaed388 (origin/exam-q1, exam-q1) Local edit conflictingg with remote
8cf79ce Update README.md
57b5f4c Local edit conflicting with remote
3783f00 Update README.md
40b2df3 Local change for conflict test
0962cba Resolved merge conflict
c01523a Local change for conflict test
b38f131 Update README.md
2acb734 (origin/test-force, origin/main, origin/HEAD, test-force, main) Revert "Ad
y text file"
f4965be Added temporary text file
0f65170 Resolved merge conflict in README.md
73efa37 Removed tracked textfiles directory
84511cb Added .gitignore to ignore textfiles directory
395154d Added textfiles directory with three files
40d6926 Update README.md
```

4. Perform a soft reset (git reset --soft HEAD~1) and observe your file and history.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q4)
$ git reset --soft HEAD~1
```

5. Make commit again.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q4)
$ git commit -m "Recommitted second change after soft reset"
[exam-q4 5b24193] Recommitted second change after soft reset
1 file changed, 3 insertions(+), 1 deletion(-)
```

6. Perform a hard reset (git reset --hard HEAD~1) and observe the changes.

```
malik@DESKTOP-N4U5EUV MINGW64 ~/lab3 (exam-q4)
$ git reset --hard HEAD~1
HEAD is now at c9657f3 First change for Q4
```
